

How to Create and Give a Technical Talk

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Outline

Logical structure

Slides

Presenting

First steps

- ▶ work out the logical structure first, not the slides
 - ▶ triage: what are the (few) main points you want to make?
 - ▶ know your audience (background, knowledge, interest . . .)
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- ▶ a good first step: explain the main points briefly and at a high level to a friend or colleague

Main components

- ▶ description of the setting and problem informally in English
- ▶ technical description of the problem
- ▶ (optionally) previous or current methods
- ▶ your method
- ▶ evaluations, comparisons to simple baseline, straw-man, or current methods

never

- ▶ mix these parts
- ▶ start with slides, clip art, format, fonts . . .

Technical problem description

describe or explain

- ▶ quantities you can choose, act on, or decide
- ▶ information or data you will use to do this
- ▶ quantities you care about
- ▶ models / equations that connect your action to what you care about
- ▶ metrics or objectives, and possibly constraints
- ▶ testing method (e.g., out-of-sample validation, back test, high fidelity simulation, ...)

Technical problem description — rules

- ▶ define every symbol or quantity that you mention
- ▶ clearly state which you can choose and which are given
- ▶ disclose model approximations and simplifications

never

- ▶ mix discussion of the problem and methods
- ▶ mix discussion of the evaluation and methods

Evaluation

- ▶ your evaluation **must** be agnostic as to method
- ▶ you will likely have several metrics or objectives
- ▶ if possible compare to baseline or current method

Evaluation methods

- ▶ out-of-sample or cross validation
- ▶ back-tests with historical data
- ▶ simulation with synthetic data
- ▶ Turing test — can you fool a domain expert?
(often but not always named Bob, who was doing this before you were born, *etc.*)

Linear versus inverted talk structure

- ▶ **linear:** background, problem, method, results
 - ▶ this should be the default structure
- ▶ **inverted:** background, results; then explain details, and how you achieved those results
 - ▶ the punch line is early
 - ▶ those who don't care about how you did it can check out

Intellectual triage

- ▶ cut talk to the bare minimum needed to get your main points across
- ▶ no, the audience does not want to hear the details of the 37 other methods you tried
- ▶ simplify the presentation, then later mention that the method can be extended to handle other settings

Example — A prediction model you've developed

you must describe

- ▶ what you're predicting, using what data
- ▶ the data you use to fit and evaluate your predictor
- ▶ the details of features and feature engineering
 - ▶ how many real features? do you standardize? take log?
 - ▶ how many categorical features? how do you represent them?
 - ▶ missing data?
- ▶ the performance metrics (e.g., error rates, ...)
- ▶ your method (optionally, baseline method)
- ▶ the evaluation method (e.g., 5-fold cross-validation)
- ▶ the final results

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General rules

- ▶ use sections that follow the logical design
- ▶ shoot for 3–5 sections
- ▶ each slide should have
 - ▶ a title
 - ▶ a clear and coherent point
 - ▶ a clear role in the overall story
- ▶ don't cram multiple topics on one slide, just because it fits
- ▶ trim the slides to the minimum possible
- ▶ in a talk you have license to simplify or even lie (a little) provided it improves clarity, pedagogy, *etc.*

Misdemeanors

- ▶ inconsistent capitalization, punctuation
 - ▶ inconsistent use of names, abbreviations, symbols
 - ▶ inconsistent mixing of telegraphic speech, full sentences
 - ▶ non-parallel bullets
 - ▶ clip art, emojis, gratuitous figures that aren't explained, . . .
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- ▶ Mykel and I (and others) **will** judge you if you commit these
 - ▶ choose a scheme and stick to it

Tips

- ▶ never refer to 'my methods', 'my results', *etc.*
(be modest, acknowledge input from others, prior work . . .)
- ▶ avoid and / or simplify equations
- ▶ don't number equations as you might in a paper

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Before the presentation

- ▶ practice at least once to check timing and flow
- ▶ do not practice more than twice
- ▶ try to imagine what you sound like to your audience (or to multiple representatives of audience types)
- ▶ try to guess some of the most likely confusions and questions, and think ahead how you'd respond

During the presentation

- ▶ don't just read the slides — use them as the scaffold
- ▶ modulate your speech; don't present in a monotone
- ▶ sign post — occasionally tell the audience where we are, why we're covering what we're covering, *etc.*
- ▶ summarize briefly at beginning and end of each section
- ▶ make observations not on the slides, e.g.,
 - ▶ that something should be surprising (or expected)
 - ▶ that a naïve sounding method actually works well in practice
 - ▶ that what you just said isn't quite true

Adapting the presentation in real-time

- ▶ adapt to audience boredom, confusion, shock, disbelief, . . .
- ▶ if you're running out of time
 - ▶ step up a level to get main ideas across
 - ▶ don't speed up (except maybe a bit)

Responding to questions

- ▶ only occasionally is a question really just a question
- ▶ instead use every question as data for **diagnostic analysis**
 - ▶ what misconception(s) would cause someone to ask that question?
 - ▶ then gently address the core misconceptions, not the original question
- ▶ a 'dumb' question isn't the asker's fault
 - ▶ it's yours for not being clear enough, or not emphasizing it
 - ▶ yes, even if it's stated very clearly on slide 6

After the talk

- ▶ post-mortem — what went well? poorly?
- ▶ maybe make changes based on questions / confusions that arose

Summary

- ▶ these are generic suggestions, which you can adapt (and occasionally violate)
- ▶ logical clarity and structure are critical; font choice is not
- ▶ recommendation:
 - ▶ first learn to be an editor or critic
 - ▶ then apply it to your own talk